Code No: 131AK **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech I Year I Semester Examinations, March/April - 2023
BASIC ELECTRICAL AND ELECTRONICS ENGINEERING
(Common to EEE, ECE, CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a) An Electric iron is rated 1000W, 240V. Find the current drawn and resistance of the heating element [2]
b) Explain the concept of power factor in a.c circuits. [3]
c) State Reciprocity Theorem. [2]
d) State and explain Superposition theorem. [3]
e) Explain PN junction diode under reverse bias condition. [2]
f) What is the purpose of filter? [3]
g) Define amplification factor (beta)? [2]
h) What do you understand by thermal runaway? [3]
i) Sketch the basic structure of a P-Channel JFET. [2]
j) Compare BJT with FET. [3]

PART - B

(50 Marks)

2.a) Obtain the equations for Star to Delta and Delta to star transformations.
b) Fin equivalent resistance across X-Y terminals for the circuit shown below Figure 1. [4+6]

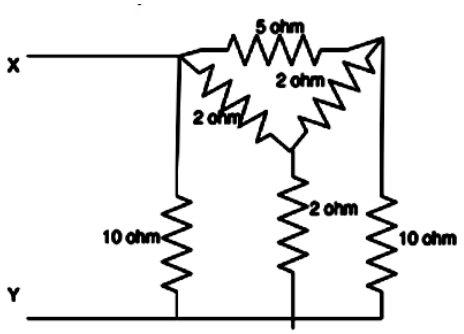


Figure 1

OR

3.a) Define the following:
i) Impedance ii) Admittance iii) Reactance iv) Susceptance

b) A coil, of resistance 35 ohms and inductance 0.02 H, carries a current of 326.5 mA, when connected to a 400 Hz a.c. supply. Determine the supply voltage. [5+5]

4.a) Find the value of R_L for maximum power transfer in the circuit Figure 2.

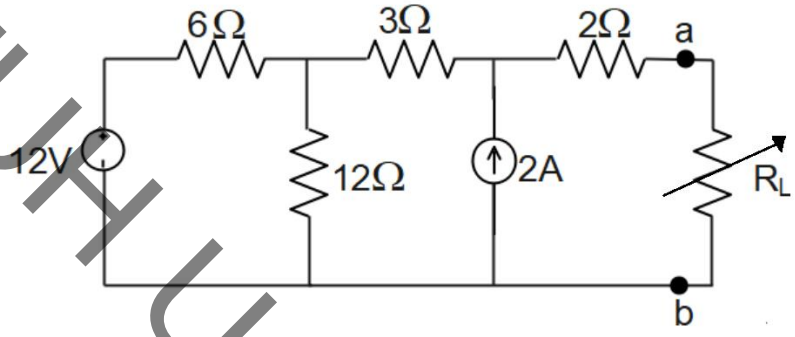


Figure 2

b) Derive the resonance frequency and quality factor for parallel R-L- C resonance circuit. [5+5]

OR

5.a) Using Norton's theorem, find the current passing through 6 Kilo ohms resistor Figure 3 .

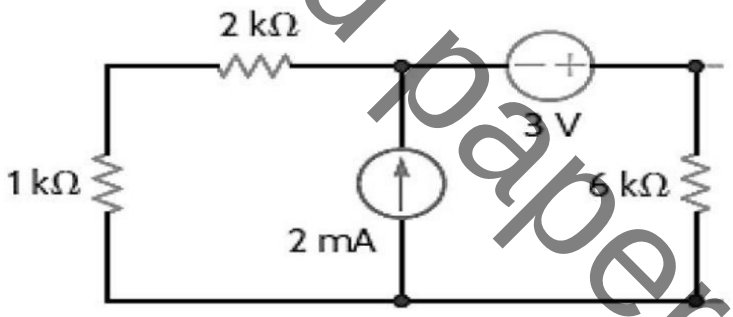


Figure 3

b) A coil of resistance 10 ohms, and inductance 1.013 H, is connected in series with a 10 μ F capacitor. Calculate (i) the resonant frequency, (ii) the circuit current, when connected to a 240 V, 50 Hz supply, and (iii) the p.d. developed across the capacitor under resonance. [5+5]

6.a) Explain the principal operation of diode and write the diode equation.

b) Draw the circuit diagram of a full-wave bridge rectifier circuit and calculate i) I_{dc} ii) $I_{r.m.s}$ iii) ripple factor iv) efficiency of rectification and v) PIV rating of diode. [5+5]

OR

7.a) Draw the Energy band diagram of PN junction diode and explain.

b) What are advantages of pi filter over capacitor filter? [5+5]

8.a) Discuss the operation of PNP transistor with diagram.

b) Explain in detail the transistor working as a amplifier. [5+5]

OR

9.a) Explain the method of evaluating h-parameters for a transistor CE configuration from its characteristics

b) What is the necessity of biasing circuits? Explain bias compensation using diodes and transistors. [5+5]

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10.a) Explain about the JFET and draw the construction of JFET.
b) Define the JFET Volt-Ampere Characteristics and determine FET parameters. [5+5]

OR

11.a) Write JFET applications.
b) Explain biasing FET. [5+5]

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